

# SRMA Workflow SOP

## How an IHSAAN Medics review actually works, from first idea to published paper

|           |                                                                                                                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document  | G3 — Core Programme Governance Pack                                                                                                                                                                                                       |
| Status    | <div>[CANONICAL]</div> · the operational backbone. The student Project Workflow Guide (ST4) is a friendlier view of this document and must point back to it, never restate it.                                                            |
| Version   | 1.0 (draft) · 29 June 2026                                                                                                                                                                                                                |
| Read with | G2 Authorship & Integrity Policy · G4 Topic Proposal & Feasibility Form · G5 Pre-Submission QA Checklist · G6 Data Management & Ownership · ST5 Team Roles · ST7 Manuscript Template · ST8 Contribution Log · S4 Supervisor Review Rubric |

### 0. What this document is

This is the single path **every** IHSAAN Medics review follows. If you ever wonder "what happens next?", the answer is here.

A **systematic review (SR)** answers one focused clinical question by finding *every* relevant study, appraising it, and bringing the findings together. A **meta-analysis (MA)** is the optional statistical step that pools numerical results from those studies into a single combined estimate. Not every review can or should do a meta-analysis — see Step 16. Together these are referred to as an **SRMA**.

You do not need ethics approval for a review of already-published literature (confirm with your own institution), you do not need funding, and you do not need to touch a single patient. That is exactly why this is the right first research project — and why it's realistic to take a beginner all the way to a publishable paper.

**What we promise you'll finish with:** a publicly registered protocol and a complete manuscript draft. Submission and acceptance are the goal, but they often continue past the 16-week programme — that is normal and is not a failure.

### 1. How we run reviews here

*This section is the calibration standard. It is the canonical home for "how we work" — G7 and S6 should point here, not repeat it. Every supervisor, regardless of experience, holds every team to these eight rules.*

1. **Register before you screen.** Every review is prospectively registered (Step 7) *before* anyone starts screening. Registering late is flagged and weakens the paper.
  2. **Two people, independently, at every judgement step.** Screening and extraction are always done by two people who decide separately, then compare. A named third person breaks ties. This is not optional — it is what makes the review valid, and it's why teams are built in small groups.
  3. **Searches are reproducible and library-supported.** If someone can't re-run your search and get your numbers, it isn't finished. Build it with a librarian (Step 6).
  4. **Narrative synthesis is a legitimate result.** We never force a meta-analysis. A well-conducted narrative synthesis is publishable and respected (Step 16).
  5. **Nothing is submitted without the QA checklist (G5) and supervisor sign-off.** No exceptions.
  6. **Authorship follows ICMJE (G2); the contribution log (ST8) is the evidence.** Doing the work is how you earn your place on the paper.
  7. **Keep records of everything.** Search strategies, screening logs, extraction sheets, decisions — all saved in the team workspace per G6.
  8. **We work to recognised standards:** PROSPERO for registration, PRISMA 2020 for reporting, RoB 2 / ROBINS-I for risk of bias, and GRADE for certainty of evidence.
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## 2. Who signs off, and when

Most day-to-day decisions are made by the team. Six decisions need **supervisor sign-off** — these are the "gates." A project does not move past a gate until the supervisor approves. Roles below are defined in full in ST5.

| Gate   | What's approved                                  | Happens at     |
|--------|--------------------------------------------------|----------------|
| Gate 1 | Topic is feasible and fills a real gap           | End of Step 2  |
| Gate 2 | Question, eligibility criteria, and protocol     | End of Step 6  |
| Gate 3 | Final search strategy (before it's run)          | Step 6         |
| Gate 4 | Synthesis direction (meta-analysis or narrative) | End of Step 16 |
| Gate 5 | Manuscript is ready in principle                 | Step 19        |
| Gate 6 | Final approval to submit (all authors)           | Step 21        |

**Team roles that own each step:** project lead / first author, deputy lead, search lead, screening lead, extraction lead, risk-of-bias lead, manuscript coordinator, referencing & submission lead, and the supervisor / senior author. See ST5.

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### 3. The pathway at a glance

| Phase                     | Weeks | Steps                                                                                        | Phase deliverable                                                   |
|---------------------------|-------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1 — Foundations           | 1-2   | Onboarding · Topic proposal · Scoping search                                                 | Approved, feasible topic (Gate 1)                                   |
| 2 — Question & Protocol   | 3-5   | PICO · Eligibility · Search strategy · Protocol · Registration                               | Registered protocol + search strategy (Gates 2 & 3)                 |
| 3 — Screening & Synthesis | 6-10  | Searching · Dedup · Screening · Flow diagram · Extraction · Risk of bias · Synthesis · GRADE | Completed analysis: flow diagram, extraction, RoB, results (Gate 4) |
| 4 — Writing & Submission  | 11-16 | Writing · Supervisor review · Journal choice · QA · Submission · Reviewer response           | Complete manuscript → submission (Gates 5 & 6)                      |

## PHASE 1 — Foundations (Weeks 1-2)

### Step 1 — Onboarding & setup

- **What it is:** Getting ready before any research starts.
- **What you do:** Read the Student Handbook (ST1), sign the Commitment Agreement (ST2), meet your team and supervisor, set up your workspace (G6), and create your accounts: a reference manager (Zotero — free), a screening tool (Rayyan — free, or Covidence if available), and your databases.
- **Tool / template:** Team workspace (Notion/Drive) · Contribution Log (ST8) started today and updated weekly from here on.
- **Signs off:** Project lead confirms the team is set up.

### Step 2 — Topic proposal & scoping search

- **What it is:** Choosing a question worth answering and checking nobody has just answered it.
- **What you do:** Draft a one-paragraph topic idea. Then run a quick **scoping search**: search **PROSPERO** for registered reviews on your topic, and **PubMed** for already-published reviews. If several recent reviews already cover it, change the question — a duplicate review is very hard to publish. Write a short feasibility note: is the question answerable from published studies, is it clinically relevant, is it narrow enough, and is there likely to be enough primary research?

- **Tool / template:** Topic Proposal & Feasibility Form (G4) · PROSPERO website · PubMed.
  - **Signs off: Supervisor — Gate 1.** They confirm the question is sound, relevant, and not already done.
  - **Common mistakes:** Question far too broad ("does exercise help health?"); skipping the redundancy check and discovering an identical review at the writing stage.
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## PHASE 2 — Question & Protocol (Weeks 3-5)

### Step 3 — Frame the question (PICO / PECO)

- **What it is:** Turning your idea into a precise, searchable question.
- **What you do:** Break the question into its parts. **PICO** = Population, Intervention, Comparator, Outcome (use **PECO**, with Exposure, for non-treatment questions). Every later step flows from this.
- **Tool / template:** PICO worksheet (section 1 of the Protocol Template).
- **Signs off:** Supervisor (rolled into Gate 2).

### Step 4 — Eligibility criteria

- **What it is:** The exact rules for which studies get in and which are excluded.
- **What you do:** Write your inclusion and exclusion criteria — study designs, population, dates, language, settings, outcomes. These must be decided *now*, before you see the results, so your choices can't be influenced by what you find.
- **Tool / template:** Inclusion/exclusion table in the Protocol Template.
- **Signs off:** Supervisor (rolled into Gate 2).
- **Common mistakes:** Vague criteria that two screeners interpret differently; deciding criteria after screening has started.

### Step 5 — Build the search strategy

- **What it is:** The detailed, repeatable recipe for finding every relevant study.
- **What you do:** With the **search lead**, build a strategy combining subject headings (e.g. MeSH terms) and free-text words with AND/OR logic, across the main medical databases — typically **MEDLINE/PubMed, Embase, and the Cochrane CENTRAL register**. Crucially, **ask a librarian / information specialist to peer-review it** — most UK medical schools and NHS trusts offer this free, and it's where weak reviews are rescued. Record the full strategy for every database so it can be re-run exactly.
- **Tool / template:** Search Strategy Log · institutional library service.
- **Signs off: Supervisor — Gate 3** (after librarian review), before the search is run.
- **Common mistakes:** Searching one database only; not saving the exact search terms; never involving a librarian.

## Step 6 — Write the protocol (PRISMA-P)

- **What it is:** Your written plan, agreed before you start, that says exactly what you'll do.
- **What you do:** Assemble Steps 3–5 into a full protocol following the **PRISMA-P** guideline (the reporting standard for review *protocols*). It states your question, criteria, search, and how you'll screen, extract, assess bias, and synthesise.
- **Tool / template:** PRISMA-P Protocol Template.
- **Signs off: Supervisor — Gate 2.** Approving the protocol approves Steps 3–6 together.

## Step 7 — Register the review (PROSPERO)

- **What it is:** Publicly recording your plan so the world knows what you set out to do — your transparency stamp.
  - **What you do:** Once the protocol is signed off and **before you screen anything**, register on **PROSPERO** (free, run by the University of York). You'll receive a registration ID (format CRD42XXXXXXX) to quote in your paper. **Important:** PROSPERO only accepts genuine systematic reviews with a health-related outcome — it will **not** accept scoping reviews or plain literature reviews. If your project is one of those, it isn't an SRMA; speak to your supervisor about either reshaping it or registering elsewhere (e.g. OSF).
  - **Tool / template:** PROSPERO website.
  - **Signs off:** Supervisor confirms before the registration is submitted.
  - **Milestone:** This is your first promised deliverable — a **registered protocol**.
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# PHASE 3 — Screening & Synthesis (Weeks 6–10)

## Step 8 — Run the search

- **What it is:** Executing the approved strategy.
- **What you do:** The search lead runs the finalised strategy across every database on the **same day where possible**, records the number of results and the date for each database, and exports everything (as RIS files) into the reference manager.
- **Tool / template:** Databases · Zotero/EndNote · Search Strategy Log (record the hit counts).
- **Signs off:** Search lead; deputy double-checks the counts.

## Step 9 — Deduplicate

- **What it is:** Removing the same study found in more than one database.
- **What you do:** Merge all results and remove duplicates, recording how many you removed (this number goes in the flow diagram).
- **Tool / template:** Zotero/EndNote, or the dedup function in Covidence/Rayyan.

## Step 10 — Title & abstract screening (dual, independent)

- **What it is:** The first sift — deciding from titles and abstracts which studies *might* be relevant.
- **What you do:** Two screeners review every record independently, marking include/exclude against the eligibility criteria, without seeing each other's decisions. Then compare. Disagreements go to the **named third reviewer** (deputy or supervisor). Be generous here — if unsure, keep it for the next round.
- **Tool / template:** Rayyan or Covidence (both log every decision automatically).
- **Signs off:** Screening lead oversees; conflicts escalate to the third reviewer.

## Step 11 — Full-text screening (dual, independent)

- **What it is:** Reading the full papers of everything that survived Step 10 and making the final include/exclude call.
- **What you do:** Retrieve the full texts. Two reviewers independently assess each one. **For every full text you exclude, record the reason** (e.g. "wrong population", "no relevant outcome") — you must report these in the paper. Conflicts go to the third reviewer.
- **Tool / template:** Rayyan/Covidence · institutional full-text access (OpenAthens).
- **Signs off:** Screening lead; third reviewer on conflicts.
- **Common mistakes:** Forgetting to log exclusion reasons (this stalls the whole write-up later).

## Step 12 — PRISMA 2020 flow diagram

- **What it is:** The standard diagram showing how you went from thousands of records to your final included studies.
- **What you do:** Fill in the numbers from Steps 8-11: records found, duplicates removed, screened, excluded (with reasons), and finally included.
- **Tool / template:** PRISMA 2020 flow-diagram generator (the free PRISMA Shiny app or the template).

## Step 13 — Data extraction (dual / checked)

- **What it is:** Pulling the key information out of each included study in a consistent way.
- **What you do:** **Pilot** the extraction form on 2-3 studies first, then extract from all included studies. Use either two independent extractors, or one extractor whose work is fully checked by a second person. Capture study details, population, intervention/exposure, outcomes, and the numbers you'll need for synthesis.
- **Tool / template:** Data Extraction Sheet (piloted) · Covidence or a structured spreadsheet.
- **Signs off:** Extraction lead; second person checks.

## Step 14 — Risk-of-bias assessment

- **What it is:** Judging how trustworthy each included study is.
- **What you do:** Two assessors independently appraise each study with the correct tool for its design — **RoB 2** for randomised controlled trials, **ROBINS-I** for non-randomised studies of interventions. (Other designs use other tools — e.g. QUADAS-2 for diagnostic accuracy; agree the right one with your supervisor.) Compare and resolve as before.
- **Tool / template:** RoB 2 / ROBINS-I templates · **robvis** for the traffic-light figure.
- **Signs off:** Risk-of-bias lead; supervisor sanity-checks the judgements.

## Step 15 — (merged into 14)

## Step 16 — Choose and run the synthesis

- **What it is:** Bringing the findings together — the heart of the review.
- **What you do:** Decide *with your supervisor* whether the studies are similar enough (in population, design, and outcomes) to pool statistically.
  - **If yes → meta-analysis:** choose the effect measure and model (usually random-effects), produce a forest plot, and assess heterogeneity ( $I^2$ ).
  - **If no → narrative synthesis:** structure the findings systematically following the **SWiM** guideline. *This is a valid, publishable result — a forced meta-analysis of dissimilar studies is worse than an honest narrative one.*
- **Tool / template:** RevMan or R (**meta**) / (**metafor**) for meta-analysis · SWiM structure for narrative.
- **Signs off:** Supervisor — Gate 4.

## Step 17 — GRADE the certainty of evidence

- **What it is:** Rating how confident readers can be in each main finding.
- **What you do:** For each key outcome, rate the certainty as high, moderate, low, or very low, and build a Summary of Findings table.
- **Tool / template:** GRADEpro GDT · Summary of Findings table.
- **Milestone:** Your analysis is now complete — flow diagram, extraction, risk of bias, results, and certainty.

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## PHASE 4 — Writing & Submission (Weeks 11-16)

### Step 18 — Write the manuscript

- **What it is:** Turning everything into the paper.

- **What you do:** The manuscript coordinator assembles the draft using the Manuscript Template (ST7), with the team writing assigned sections. Follow the **PRISMA 2020 checklist** as you write — it tells you exactly what each section must report. Keep the PROSPERO ID, full search strategy, and PRISMA checklist ready as supplementary files.
- **Tool / template:** Manuscript Template (ST7) · PRISMA 2020 checklist.
- **Signs off:** Manuscript coordinator assembles; team drafts.
- **Milestone:** This is your second promised deliverable — a **complete manuscript draft**.

### Step 19 — Supervisor critical review

- **What it is:** Senior eyes on the whole paper before it goes anywhere.
- **What you do:** The supervisor reviews the draft against the Supervisor Review Rubric (S4) and returns major/minor revisions. Expect at least one or two rounds. This is the supervisor's core authorship contribution — genuine critical revision, not a rubber stamp.
- **Tool / template:** Supervisor Review Rubric (S4).
- **Signs off:** **Supervisor — Gate 5** when the paper is ready in principle.

### Step 20 — Choose a journal (and screen out predators)

- **What it is:** Finding a legitimate, well-matched home for the paper.
- **What you do:** Shortlist journals whose scope fits and whose readership is right. **Filter out predatory journals:** run each candidate through **Think-Check-Submit**, confirm open-access titles are listed in the **DOAJ**, check it's indexed in PubMed/MEDLINE, and check any article processing charge (and whether waivers exist). When unsure, ask the supervisor.
- **Tool / template:** Think-Check-Submit · DOAJ · journal scope/author guidelines.
- **Signs off:** Supervisor approves the target journal.
- **Common mistakes:** Being lured by a flattering email promising fast publication for a fee — a classic predatory-journal trap that students are targeted with.

### Step 21 — Pre-submission QA & author approval

- **What it is:** The final safety net.
- **What you do:** Run the whole paper through the **Pre-Submission QA Checklist (G5)**: reproducible search, dual screening done, exclusion reasons recorded, extraction checked, right RoB tool, PRISMA checklist complete, references accurate (and real — verify every one), claims matched to the evidence, journal legitimate. Then **every author confirms they meet ICMJE criteria (G2) and approves the final version**.
- **Tool / template:** QA Checklist (G5) · Authorship & Integrity Policy (G2) · Contribution Log (ST8).
- **Signs off:** **Gate 6 — all authors**.



- **What it is:** Sending the paper to the journal.
- **What you do:** The corresponding/senior author (usually the supervisor) submits or directly supervises submission: cover letter, formatted files, supplementary search strategy and PRISMA checklist, and the PROSPERO ID. **Update the PROSPERO record** to mark the review as submitted (and later add the DOI once published).
- **Tool / template:** Journal submission system · PROSPERO record.
- **Signs off:** Corresponding/senior author.

## Step 23 — Respond to reviewers

- **What it is:** Working with the journal's feedback to get to acceptance.
  - **What you do:** When reviews come back, write a polite, **point-by-point response letter**, make the agreed changes (tracked), and resubmit. The supervisor oversees this. Revisions can take weeks to months — this is the normal road to a published paper, and it usually continues beyond Week 16.
  - **Tool / template:** Response-to-reviewers letter · tracked-changes manuscript.
  - **Signs off:** Supervisor oversees each revision round.
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## 4. A note on records (see G6)

Throughout, save everything in the team workspace: the exact search strategies and hit counts, the screening exports, the piloted extraction sheet, the risk-of-bias judgements, and every version of the protocol and manuscript. Journals increasingly ask for these, and they are what makes your review reproducible. Keep your Contribution Log (ST8) up to date weekly — it is the evidence base for authorship decisions.

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## 5. Mini-glossary

- **Systematic review (SR):** a structured review that finds and appraises all studies on one question.
- **Meta-analysis (MA):** the optional statistics step that pools study results into one combined estimate.
- **PICO / PECO:** a way of structuring your question — Population, Intervention (or Exposure), Comparator, Outcome.
- **PROSPERO:** the free international register where you record your review plan in advance.
- **PRISMA 2020:** the standard checklist and flow diagram for *reporting* a systematic review.
- **PRISMA-P:** the reporting standard for the *protocol*.

- **RoB 2 / ROBINS-I:** tools for judging the risk of bias in trials / non-randomised studies.
  - **GRADE:** a system for rating how certain we can be in each finding.
  - **Narrative synthesis (SWiM):** structured written synthesis used when studies can't be pooled — a valid result.
  - **Predatory journal:** a journal that charges a fee but skips proper peer review; avoided using Think-Check-Submit and the DOAJ.
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